Bio-303

Bio Chemistry II

- Ligand:

In biochemistry and pharmacology, a ligand is a substance that forms a complex with a biomolecule to serve a biomolecule to serve a biological purpose. In a narrow sense, it is a single triggering molecule binding to a site on a target protein. A molecule bound reversible by a protein is called Ligand. Ligands include substrate, inhibitor, activator. A Ligand binds at a site on the protein called the binding site, binding site is complementary to the Ligand in size, shape, charge and hydrophobic or hydrophilic character.

Classification of Enzymes:

Enzymes are classified into 6 groups.

- Oxidoreductase.
- Hydrolases.
- Isomerases.
- Lyases.
- Transferases.
- Ligases.

i- Oxidoreductases:

An Enzyme which catalyze oxidation/reduction reaction are called oxidoreductases. E.g.

Cytochrome oxidase oxidize cytochromes.

ii- Transferases:

An Enzyme which catalyze the transfer of functional group is called transferases. E.g.

Transaminase or haxokinase thransfers the phosphate group from ATP to glucose.

iii- Hydrolysas:

The enzyme which bring about the ~~large~~ break down of large complex organic molecules into smaller ones by adding water and break down specific covalent bonds E.g.

Lipases Lipid hydrolyzing Enzymes, Peptidases Protein hydrolyzing Enzyme.

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tRNA → 74-95 nucleotides → smallest → compose 20% of total cellular RNA.
rRNA → → 80% of the total RNA in cell.
mRNA → 500 – 6000 Nucleotides → 2% to 5% of cellular RNA.

iv- Lyases:
the enzymes which which catalyze the breakdown of specific covalent bond and removal of groups without ~~addin~~ hydrolysis. E.g.
Histidine decarboxylase breaks specific covalent bonds b/w histidine & carbon atom forming CO_2 & histamine.

v- Isomerase:
The enzyme which bring about intramolecular rearrangement of atoms in the molecules & thus forming isomers. E.g.
phosphohexose isomerase change glucose-6-phosphate into fructose-6-phosphate.

vi- Ligases:
The enzyme which bring about the joining of two different molecules.
polymerase are that join monomers to form polymers i-e. DNA or RNA.

Small Nuclear RNA:
Small Nuclear RNA are large number of small stable RNA species found in Eukaryotic cells. Most of them are complexed with proteins to form ribonucleoproteins. they are distributed in nucleus, cytoplasm or in both. they are involve in rRNA and mRNA processing & in gene regulating.

Pentose Phosphate Pathway:
The pentose phosphate pathway also called phosphogluconate pathway and hexos monophosphate shunt is a metabolic pathway parallel to glycolysis. It generate NADPH and pentoses as well as ribose 5-phosphate a precursor for the synthesis of nucleotide. while the pentose phosphate pathway does involve oxidation of glucose, its primary role in anabolic rather than catabolic.

Steps involved:

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i- Oxidative ii- Non-Oxidative

i- Oxidative phase:

~~Glucose-6-phosphate is~~

i- In oxidative stage, two molecules of NADP+ are reduced to NADPH utilizing the energy from the conversion of glucose-6-phosphate into ribulose-5-phosphate. These NADPH molecules can be then be used as an energy source in the cell.

ii- Next a carbon is removed and CO_2 start subscript, 2, end subscript is released.

ii- Non-oxidative phase:

The non-oxidative phase of pentose phosphate pathway involves the conversion of ribulose-5-phosphate to ribose-5-phosphate through a series of independent reactions. In this phase no NADPH molecule is created. In this phase ~~Ribose~~ Ribulose-5-phosphate can be converted into two different 5-carbon molecules.

Glycolysis:

Glycolysis is a metabolic pathway that convert glucose into pyruvic acid. The free energy is released in this reaction is used to form high energy molecules ATP and reduce nicotinamide adenine dinucleotide. Glycolysis is a sequence of 10 reactions catalyze by enzymes.

Steps in Glycolysis:

Glycolysis involve 10 steps:

i- In the first step of glycolysis glucose is phosphorylated to glucose 6-phosphate this step is catalyze by hexokinase.

ii- In the second step of glycolysis the isomerization of glucose-6-phosphate, an aldose to fructose-6-phosphate, an ketose occurs which is catalyzed by phosphoglucose isomerase.

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3- In the third step of glycolysis a second phosphorylation reaction occurs in which fructose-6-phosphate is converted into fructose 1,6 biphosphate which is catalyze by Phosphofructokinase 1.

4- In the forth step fructose 1,6-bisphosphate is converted into glyceraldehyde 3-phosphate which is catalyze by aldolas or 1-6 bisphospha aldolase.

5- In the fifth step interconversion of dihydroxyacetone phosphate and glyceraldehyde 3-phosphate occurs - this reaction is catalyze by triose phosphate isomerase.

6- In the Sixth step oxidation of glyceraldehyde 3-phosphate into 1,3-bisphosphoglycerate occur which is catalyze by 3-phosphate de hydrogenase.

7- In the seventh step 3-phosphoglycerate and a molecule of ATP form this reaction is catalyze by phosphoglycerate kinase.

8- In the eight step 3-phosphoglycerate is converted into 2-phosphoglycerate this reaction is catalyze in the presence of phosphoglycerate mutase.

9- In the ninth step 2-phosphoglycerate is dehydrated to form phospho enolpyruvate in a reversible reaction which is catalyze by enolase

10- In the final step of glycolysis the transfer of phosphoryl group from phosphoenolpyruvate to ADP to form pyruvate and ATP - this reaction is catalyze by pyruvate kinases.

The enzyme Transketolase use what cofactor in Pentose Phosphate pathway?

the cofactor used is thiamine pyrophosphate.

In Glycolysis, three conversion reactions are noteworthy write those?

one molecule of Glucose, a six carbon sugar and 2 molecules of pyruvate

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What is half life ?

half life in radioactivity is the interval of time required for one-half of the atomic nuclei of a radioactive sample to decay.

Example: The radioactive isotope cobalt-60 which is used for radio ~~activ~~ therapy has a half life of 5.26 years.

Glucose transporters:

SGLT1, SGLT2, SGLT3.

NAD and NADPH abbreviation:

Nicotinamide adenine dinucleotide

Nicotinamide adenine dinucleotide phosphate.

What is acyl group ? It is derived from ? Example.

An acyl group is a moiety made by removal of one or more hydroxyl groups from an oxoacid including inorganic acid. It contains a double bonded oxygen atom and alkyl group.

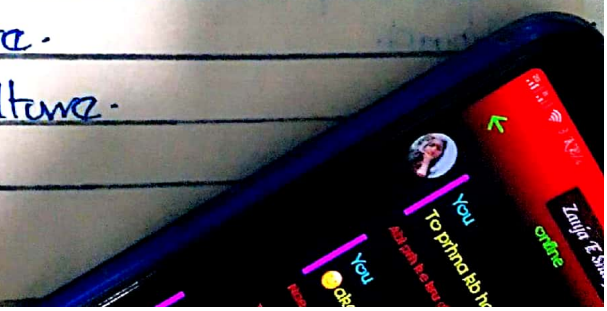
It is derived from carboxylic acid. ; Examples are Esters, aldehyde

Feed Batch Culture ?

Fed batch culture is defined as an operational technique in biotechnological processes where one or more nutrients are fed to the bioreactor during cultivation of and in which the products remain in the bioreactor until the end of the run.

Type of Fed-batch culture:

- Fixed-Volume Fed batch culture.
- Variable-Volume Fed batch culture.
- Repeated or cyclic Fed batch culture.
- Single Fed-batch culture.



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ENEDIOL :

an organic compound characterized by the grouping >C(OH)−C(OH)< containing 2 hydroxyl groups adjacent to a double bond, a reducing sugar to form enidiol. In mild alkaline solutions, carbohydrates containing a free sugar group (aldehyde or keto) will tautomerize to form enediol, where two hydroxyl are attached to the double bonded carbon.

Entropy :

Entropy is the measure of a systems thermal energy per unit temperature that is unavailable for doing useful work. the amount of entropy is also a measure of the molecular disorder, or randomness of a system. units for entropy are J/mol.K.

deficiency cause Wair Syndrome.
Thiamine B12.

Enzymes in Photosynthesis.
Rubisco, pyruvate, phosphate dikinase, NADP-malic enzyme.

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density of pure water = 1000
Molar mass : 18 g/mol
Molar concentration : 55.mol/L

Second law:
The universe always tends toward increasing disorders; in all natural processes, the entropy of the universe increase Eg. a hot cup of tea.

Enthalpy:
Enthalpy is the heat content of the reacting system.
It reflects the number and kind of chemical bonds in the reacta and products. When a chemical reaction release energy - it so to be exothermic and the value of enthalpy (H) is negative wh in case of endothermic .....

Gibbs free Energy: G
It express the amount of energy capable of doing work durin a reaction at constant temperature and pressure.
When a chemical reaction proceed with the release of energy. "G has a negative value and the reaction is said to be exergonic E in case of endergonic .....

$$>C(OH)-C(OH)<$$

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Why pure water carries Electric Current?

Water is capable of conducting electric current due to the dissolved ions and impurities. when a battery with +ive & -ive pole is placed in water, the +ive ions are attracted by the -ive pole and negative ions are attracted by +ive pole, thus creating a current.

2- Enzyme:

Enzymes are catalytic protein which are biologically active and have ability to increase the efficiency of chemical reaction by lowering the activation energy.

3- Energy of activation:

The minimum amount of energy required to initiate a chemical reaction is called activation energy.

In non-living system we use heat as a source of energy to increase the number of effective collision between molecules but in living system we can't use heat as a source of energy of activation bcz organisms and cells are made up of thermo liable i.e temp sensitive protein molecules.

In living organisms 1000 chemical reactions are taking place at a time. Energy of activation required for such a large number of reaction cannot be provided by living system.

4- Effect of Temp on Enzyme.

Rate of Enzymatic reaction increase with inc in Temp but upto certain limit - Heating increase molecular motion. thus molecule of substrate and enzyme move more quikly & increase the probability of reaction. If the temp increased above optimum

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temp, then the rate of reaction start decreasing. Due to increase of temp, secondary & tertiary structure of protein disrupted and enzyme become denature. Due to this enzyme unfold and structure of active site lost.

Optimum Temp:

The temp at which activity of enzyme is maximum.

- Effect of pH on Enzyme:

The pH at which rate of reaction is high.

Change in pH alters the ionic charge of the acidic and basic group and alters the ionic bording which helps to maintain the specific shape of enzyme.

Change in pH change the shape of enzyme and its active site. Extreme change in pH denatured the enzyme.

- Effect of Substrate concentration:-

When substrate concentration is low, the reaction rate increase with the increase in substrait concentration. If the enzyme concentration kept constant and the amount of substrate increases, a point will come when the further increase in substrate does not increase the rate of reaction anymore. this is bcz of the fact that all the active sites are occupied and no active site is available for substrate.

the increase in substrate (S) level increase the velocity (V) of the enzymatic reaction first. the reaction ultimately reaches a maximum velocity at equilibrium state. This rise in velocity V is decreased progressively with furthur increase in (S). The reaction does not increased by any further rise in substrate concentration. This happens when all the active sites are occupied.

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Enzyme Inhibition:

Chemical substances which inhibits the activity of enzymes are called inhibition.

Types:

2 types ⇒ Competitive Inhibitor, 2- Non-Competitive

Competitive Inhibitor:

The type of inhibitor which attach with the active site due to similarity of substrate is called competitive inhibitors.

- Competitive inhibitors are usually similar to substrate so fits into active site. However, not similar enough to substitute fully the normal substrate.
- Competitive inhibitor temporarily occupies the active site & does not damage the active site permanently.
- It support lock and key model.

Example include melonate.

Non-Competitive Inhibitor:

The type of inhibition which bind to enzyme but not to at active site. is called Non-Competitive inhibitor.

It Inactivates the enzymes by altering its shape so that the substrate can not bind to the active site

- It support induced fit model.
- Example includes salts of heavy metals and cyanides. are potent poisons bc2 they can kill the living organism by inhibiting cytochrome oxidase essential for cellular respiration.

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Equilibrium Constant:

For a chemical reaction, the equilibrium constant can be defined as the ratio b/w the amount of reactant and the amount of product which is used to determine chemical behaviour.

At Equilibrium State, Rate of forward reaction is equal to rate of backward reaction.

$$K_c = K_f = K_r.$$

9- Types of Chemical reactions.

There are five basic types.

i- Synthesis, ii- decomposition, iii- combustion, iv- single replacement, v- double replacement.

~~How metals obtained through light Metals in pure state?~~

Standard Reduction Potential?

When two conjugate redox pairs are ~~Red~~ together in solution, electron transfer from the electron donor of one pair to the electron acceptor of other.

The tendency for such a reaction depends on the relative affinity of the electron accepter of each redox pair for electrons, the measure of this affinity (in volts) is called standard reduction potential (E^0).

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Which form of pyruvate can immediately tautomerize? what does it form.

In the substrate lvl phosphorylation, the product of pyruvate first appears in its enol form, then tautomarize rapidly and nonenzymatically to its keto form which predominates at pH 7.

- Define pH. According to Equation on 25°C?

The pH is defined as the logarithm to the base 10 of the reciprocal of $[H^+]$ i.e the negative logarithm of $[H^+]$.

$$pH = \log 1/[H^+]$$

$$pH = -\log [H^+]$$

The pH of water at 25°C in which H^+ and OH^- ions are present in equal numbers i.e 7.0. thus,

$$pH = -\log [1 \times 10^{-7}] = 7$$

The abbreviation of pH is power of hydrogen ion.

For each pH units less than 7.0 the $[H^+]$ is increased to fold and for each pH unit above 7.0 the $[H^+]$ is decreased tenfold.

Machanism of Enzyme Action:

In catalysis substrate first bind to the active site of the enzyme to form enzyme substrate complex. later the enzyme converts substrate into product. while it is attached to the enzyme (EP complex). Enzyme regulate internal and external conditions for reaction.

$$E + S \rightarrow ES \longrightarrow EP \longrightarrow E + P.$$

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Ionization:

Ionization is the process by which an atom or a molecule aquires a negative or positive charge by gaining or losing electrons to form ions, often in conjuction with other chemical change.

Isozymes:

Two or more enzymes that catalyze the same reaction but are encoded different genes are called isozymes.
Example of Isozymes include glucokinase a variant of hexokinase.

Three major fates of animal & vascular plants:

In animal and vascular plants glucose has 3 major fates.
i- It may be stored as sucrose-
ii- Oxidize to a 3-carbon compound via glycolysis to provide ATP.
iii- Oxidize via-pentose-phosphate pathways to yield ribose-5-phosphate.

What is thioalcohol:

when oxygen atom of an alcohol is replaced with with a sulfur ato

Glyconeogenesis?

The process of synthesis of glucose or from non-Carbohydrate source is known as Glyconeogenesis. or it can also be defined as It is the pathway which convert pyruvate and related three and four carbon compounds to glucose.

$$\text{Glucose-6-phosphate} + H_2O \longrightarrow \text{Glucose} + P_i \quad (36).$$

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6 types of fermentars.

i- Fed batch fermentar, ii- Air Lift or bubble fermentar iii- Tower fermentar, iv- photo bioreactor, v- Paked bed bioreactor, vi- Membrane bioreactor.

Half Reaction:

A half reaction is either the oxidation or reduction reaction component of redox reaction. A half reaction is obtained by considering the change in oxidation state of individual substances involved in a redox reaction.

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